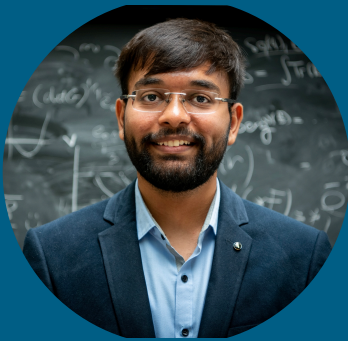


VATSALYA VAIBHAV



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Links

/GitHub
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About Me

I am a final year Ph.D. student at the Higgs Centre for Theoretical Physics at the University of Edinburgh. I am a cosmologist in spirit which means I am motivated to understand the Universe from the shortest length scales to the largest. My research includes a wide range of topics from quantum gravity and cosmology, lattice discretizations of fermions, to mathematical physics.

EDUCATION

Higgs Centre for Theoretical Physics, University of Edinburgh (Doctoral Degree)

EDINBURGH SEPTEMBER 2023 – AUGUST 2027
Degree: Doctor of Philosophy in Physics

Thesis: Fermions, in the sky, and on the lattice

Supervisor: Prof. Latham Boyle

Perimeter Institute for Theoretical Physics, University of Waterloo (Graduate Degree)

WATERLOO ONTARIO SEP 2022 – JUNE 2023
Degree: Master of Science
Perimeter Scholar International

Thesis: Quantized Supertwistors, Split-Octonions, and Unification

Supervisor: Prof. Latham Boyle

Indian Institute of Technology Kanpur (Undergraduate Degree)

KANPUR AUG 2018 – JULY 2022
Degree: Bachelor of Science (Physics)
Grade: 9 CGP

PUBLICATIONS

1. L. Boyle, **V. Vaibhav**, "A New Approach to Chiral Gauge Theories on the Lattice", to appear soon.
2. V. Nenmeli, **V. Vaibhav**, "Cosmological perturbation theory for decelerating spatially flat FRW spacetimes in the Bondi-Sachs formalism", to appear soon.
3. L. Boyle, N. Turok, **V. Vaibhav**, "Might the radiation era extend back to the Big Bang? On dark matter production and the relic graviton background in quadratic gravity", arXiv:2609.11750.
4. L. Boyle, N. Turok, **V. Vaibhav**, "Fixed points of classical gravity coupled with a Standard-Model-like theory" arXiv:2509.09346.
5. P. Kaushik, **V. Vaibhav**, T.P. Singh, "An $E_8 \otimes E_8$ unification of the standard model with pre-gravitation, on an octonion-valued twistor space", Zeitschrift für Naturforschung A, 2026. <https://doi.org/10.1515/zna-2026-0190>, arXiv:2206.06911.
6. **V. Vaibhav** and T.P. Singh, "Left-Right symmetric fermions and sterile neutrinos from complex split biquaternions and bioctonions", Adv.Appl.Clifford Algebras 33 (2023) 3, 32, arXiv: 2108.01858.
7. V. Bhatt, R. Mondal, **V Vaibhav**, and T.P. Singh, "Majorana neutrinos, exceptional Jordan algebra, and mass ratios for charged fermions" 2022 J. Phys. G: Nucl. Part. Phys. <https://doi.org/10.1088/1361-6471/ac4c91>, arXiv:2108.05787.

CONFERENCES AND TALKS

1. **New dark matter production mechanism in quadratic gravity**, poster presentation at ERGE 2026, University of Sussex.
2. **Can the radiation era extend back to the Big Bang?**, seminar talk at Kobayashi Maskawa Institute for Theoretical Physics, Nagoya.
3. **A new approach to chiral gauge theories on the lattice**, talk at Frontiers of Lattice Fermions 2026, Yukawa Institute for Theoretical Physics, Kyoto.
4. **Can the radiation era extend back to the Big Bang?**, talk at Cosmological Frontiers in Fundamental Physics 2026, Perimeter Institute for Theoretical Physics, Waterloo Canada. <https://pirsa.org/speaker/vatsalya-vaibhav-0>
5. **A gravitational theory on the Lattice**, talk at *Lattice 2025, TIFR, Mumbai*.
6. **A physicist's proof of the Atiyah-Singer Index Theorem**, Student seminar, *Higgs Centre*, 2025.
7. **Fixed points of classical gravity coupled with a standard-model-like theory**, poster presentation at *Cosmological Frontiers in Cosmology, IAP Paris, 2025*.
8. **Fixed points of classical gravity coupled with a standard-model-like theory**, Gong show talk at *Tales of Gravity, Nottingham, 2025*.
9. **Asymptotically safe cosmology**, talk at *Young Theorists Forum (YTF), IAPP Durham, 2024*.
10. **Introduction to the exact renormalization group equation**, student seminar, *Higgs Centre, 2024*.
11. **Superfun with super-composition algebras and their application to supersymmetric spacetimes and the standard model**, student seminar, *Higgs Centre, 2023*.

TEACHING

- | | |
|---------------------------------------|-------------------------|
| 1. Symmetries of Particles and Fields | (Fall 2023, 2024, 2025) |
| 2. Quantum Field Theory | (Fall 2023, 2024) |
| 3. Gauge Theories in Particle Physics | (Spring 2024, 2026) |
| 4. Classical Electrodynamics | (Spring 2025) |
| 5. Symmetries in Quantum Mechanics | (Spring 2025) |

SCHOLARSHIPS AND AWARDS

1. **Edinburgh Doctoral Scholarship**: prestigious scholarship from the University of Edinburgh
2. **Perimeter Scholars International Award**: prestigious scholarship from the University of Waterloo.
3. **National Standard Examination in Physics** state top 1% and qualified for the Indian National Physics Olympiad (**INPHO**)

VATSALYA
VAIBHAV

Mentoring

- Co-supervised a Master's project on boundary conditions for self-dual gravity at the Higgs Centre for Theoretical Physics.
- Mentored a summer reading project for first and second-year undergraduate students at IIT Kanpur on the book 'Spinors and Spacetime'.

Journal Refereeing

- The European Physical Journal C (EPJC): Particles and Fields.
- Nuclear Physics B: high energy physics, interface of particle physics and cosmology, quantum field theory, statistical systems, and mathematical physics.

Volunteering Services

- Organizer for the theoretical and mathematical physics student seminars at the Higgs Centre. 2024-2026
- Member of the core team (academics) of the institute counselling service at IIT Kanpur.